

ECE 3337: Test #2

Spring 2026

Last name of student POSTED	Student ID number SOLUTIONS
First name of student _____	Email address _____

Do This First:

1. Make sure that you have all the pages of the exam.
2. Fill in the above information.
3. Turn off and put away all electronic gadgets (laptop, tablet, smartphone, calculator, smart watch, ...)

Exam Rules:

1. You are allowed to bring one 8.5" x 11" handwritten crib sheet (double sided).
2. Indicate your answers in the spaces provided. Use the back of the page to carry out calculations.

Q1	Q2	Q3	Q4	Q5	Q6	Total
20	20	25	15	10	10	100
Laplace Transform	Inverse Laplace Transform	Circuit Analysis	Response Partitions	Op Amp Circuits	Block Diagrams	_____
_____	_____	_____	_____	_____	_____	_____

1. (20 points) Laplace Transforms

a. (8 points) Find $X(s)$, the Laplace transform of the following signal:

$$x(t) = 3\delta(2t - 1) + u(t - 1) + t^2 e^{-2t} u(t) + t e^{-t} \sin(t) u(t).$$

$$= \frac{3}{2} \delta(t - \frac{1}{2}) + u(t - 1) + t^2 e^{-2t} u(t) + t e^{-t} \sin(t) u(t)$$

$$\Rightarrow X(s) = \underbrace{\frac{3}{2} e^{-\frac{s}{2}}}_{(A)} + \underbrace{\frac{e^{-s}}{s}}_{(B)} + \underbrace{\frac{2}{(s+2)^3}}_{(C)} - \frac{d}{ds} \left\{ \underbrace{\frac{1}{(s+1)^2 + 1}}_{\substack{\uparrow \\ \frac{2(s+1)}{[(s+1)^2 + 1]^2}} (D)} \right\}$$

Grading: • 1 pt attempt
• +2 pts each term

b. (4 points) Given the signal $x(t)$, find the Laplace transform of $y(t) = 5x(2t) + \int_{0-}^t x(\tau) d\tau$ in terms of $X(s)$.

$$Y(s) = \frac{5}{2} X\left(\frac{s}{2}\right) + \frac{X(s)}{s}$$

Grading: • 1 pt attempt (A) (B)
• 2 pts each term being correct

c. (8 points) Find the Laplace transform of $y(t) = tx(t)$ when $x(t) = \cos(t)u(t)$.

$$X(s) = \frac{s}{s^2+1} \quad \textcircled{A}$$

$$\Rightarrow Y(s) = -\frac{d}{ds} \left\{ \frac{s}{s^2+1} \right\} = \frac{2s^2}{(s^2+1)^2} - \frac{1}{(s^2+1)} \quad \textcircled{C}$$
$$= \frac{s^2-1}{(s^2+1)^2} \quad \textcircled{B}$$

Grading:

- 2 pts for attempt
- +2 pts each for $\textcircled{A}$, $\textcircled{B}$, $\textcircled{C}$

2. (20 points) Inverse Laplace Transform

Find the inverse Laplace transforms for the following systems. Show your steps clearly.

a. (10 points) $X(s) = \frac{(s+4)}{(s^2+10s+21)} e^{-6s}$.

Let $X_1(s) = \frac{(s+4)}{(s^2+10s+21)}$

$$\begin{aligned} X_1(s) &= \frac{(s+4)}{(s+7)(s+3)} \quad \text{--- (A)} \\ &= \frac{(s+3)+1}{(s+7)(s+3)} \\ &= \frac{1}{(s+7)} + \frac{1}{(s+7)(s+3)} \\ &= \frac{3}{4(s+7)} + \frac{1}{4(s+3)} \quad \text{(B) } \left. \vphantom{\frac{3}{4(s+7)} + \frac{1}{4(s+3)}} \right\} \text{(C)} \end{aligned}$$

The inverse Laplace transform of each of the terms is

$$\frac{1}{(s+7)} \xleftrightarrow{\mathcal{L}^{-1}} e^{-7t}u(t) \quad \text{(D)}$$

$$\frac{1}{(s+3)} \xleftrightarrow{\mathcal{L}^{-1}} e^{-3t}u(t) \quad \text{(E)}$$

Combining, $X_1(s) \xleftrightarrow{\mathcal{L}} x_1(t)$

$$\frac{3}{4(s+7)} + \frac{1}{4(s+3)} \xleftrightarrow{\mathcal{L}} \frac{3}{4}e^{-7t}u(t) + \frac{1}{4}e^{-3t}u(t)$$

Using the time shifting property to include the effect of e^{-6s} , (F)

$$x(t) = \frac{3}{4}e^{-7(t-6)}u(t-6) + \frac{1}{4}e^{-3(t-6)}u(t-6)$$

(G) (H)

2 points for trying. 1 point for A-H.

b. (10 points) $X(s) = \frac{s^3+s+4}{s^2+1}$.

The order of the numerator is greater than the order of the denominator. Therefore, perform long division.

$$X(s) = \frac{s^3 + s + 4}{s^2 + 1} = \frac{s(s^2 + 1) + 4}{s^2 + 1} = s + \frac{4}{s^2 + 1}$$

(A)

$$X(s) = s + 4 \frac{1}{s^2 + 1}$$

Finding the inverse of each term gives

$$1 \xleftrightarrow{\mathcal{L}^{-1}} \delta(t)$$

$$s \xleftrightarrow{\mathcal{L}^{-1}} \frac{d}{dt}(\delta(t)) = \delta'(t)$$

(B)

$$\frac{1}{s^2 + 1} \xleftrightarrow{\mathcal{L}^{-1}} \sin(t) u(t)$$

(C)

Combining

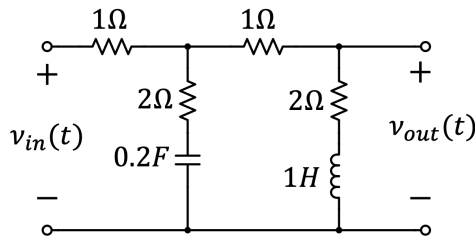
$$x(t) = \delta'(t) + 4 \sin(t) u(t)$$

(D)

2 points for trying. 2 points for A-D.

3. (25 points) Circuit Analysis

For the circuit below, the input $v_{in}(t) = 1V$ for $t < 0$ and $v_{in}(t) = te^{-2t}u(t)$ for $t > 0$.

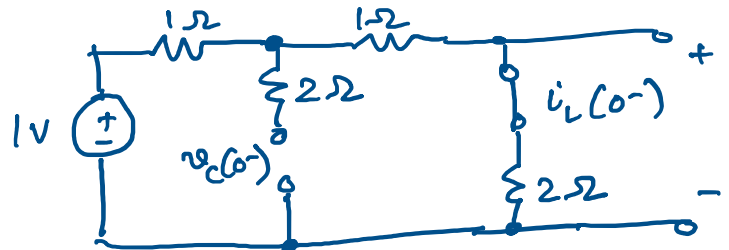


- a. (5 points) Draw the s-domain representation of the circuit with the initial conditions indicated.

Initial conditions:

$$i_L(0^-) = 0.25 A$$

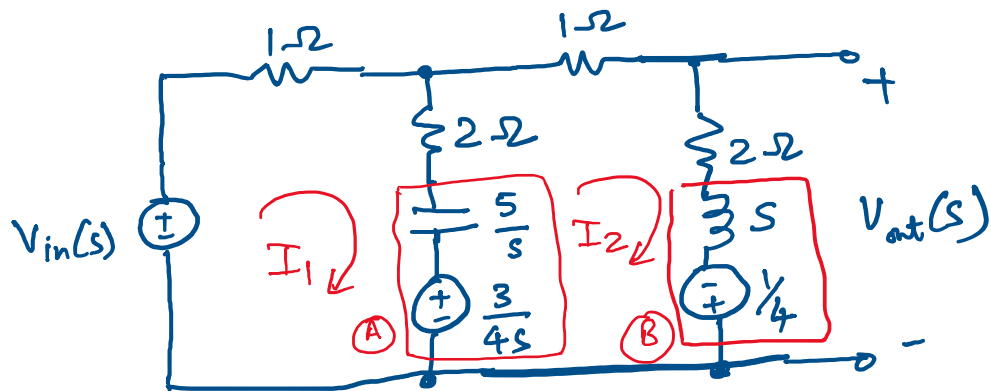
$$v_C(0^-) = 0.75 V$$



s-domain circuit

Grading: .1 pt trying

• +1 pt each (A), (B)



- b. (5 points) Find the s-domain representation of the input $v_{in}(t)$ for $t > 0$.

$$v_{in}(t) = te^{-2t}u(t)$$

$$V_{in}(s) = \frac{1}{(s+2)^2}$$

Grading: .2 pts trying

• 5 pts correct answer

c. (10 points) Write the mesh current equations in the s-domain needed to find the output $V_{out}(s)$. Express your answer by writing out the entries in the Cramer's method matrix shown below in the Laplace domain. Here, I_1 and I_2 represent the mesh currents (left to right), and V_1 , V_2 represent voltages.

$$\begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

KVL:

$$\underbrace{I_1 \left(1 + 2 + \frac{5}{s} \right)}_{A_1} - \underbrace{I_2 \left(2 + \frac{5}{s} \right)}_{A_2} = \underbrace{V_{in} - \frac{3}{4s}}_{V_1}$$

$$- \underbrace{I_1 \left(2 + \frac{5}{s} \right)}_{A_3} + \underbrace{I_2 \left(\frac{5}{s} + 2 + 1 + s + 2 \right)}_{A_4} = \underbrace{\frac{3}{4s} + \frac{1}{4}}_{V_2}$$

Grading • 2pts for attempt
• + 2pts each entry that is correct

d. (5 points) Write the formula for $V_{out}(s)$ in terms of $V_{in}(s)$. Your answer should be written in terms of A_1 , A_2 , A_3 , A_4 , I_1 , I_2 , $V_{in}(s)$, $V_{out}(s)$.

$$\textcircled{A} \quad \Delta = A_1 A_4 - A_3 A_2 \Rightarrow I_2 = \frac{\Delta I_2}{\Delta}$$

$$\textcircled{B} \quad \Delta_{I_2} = A_1 V_2 - A_3 V_1 = \frac{A_1 V_2 - A_3 V_1}{A_1 A_4 - A_3 A_2} \textcircled{C}$$

$$V_{out}(s) = (s+2) I_2 - \frac{1}{4} \textcircled{D}$$

Grading: • 1 pt for attempt + 1 pt each $\textcircled{A}$, $\textcircled{B}$, $\textcircled{C}$, $\textcircled{D}$.
• Full pts for correct $\textcircled{D}$.

4. (15 points) System Response Partitions

A linear time-invariant system is described by the following differential equation:

$$\frac{d^2 y(t)}{dt^2} + 7 \frac{dy(t)}{dt} + 10y(t) = x(t)$$

The input is $x(t) = e^{-jt}u(t)$, with the initial conditions $y(0^-) = 0$, $y'(0^-) = 2$.

- a. (5 points) Derive a formula for the output $Y(s)$. Your answer should be expressed as a ratio of polynomials in s with a fully factored denominator.

$$\begin{aligned} y(t) &\stackrel{\mathcal{L}}{\leftrightarrow} Y(s) \\ \frac{d}{dt} y(t) &\stackrel{\mathcal{L}}{\leftrightarrow} sY(s) - y(0^-) \\ \frac{d^2}{dt^2} y(t) &\stackrel{\mathcal{L}}{\leftrightarrow} s^2 Y(s) - sy(0^-) - y'(0^-) \end{aligned}$$

Therefore, taking the Laplace transform of the LCCDE yields:

$$\begin{aligned} s^2 Y(s) - sy(0^-) - y'(0^-) + 7(sY(s) - y(0^-)) + 10Y(s) &= X(s) \\ s^2 Y(s) + 7sY(s) + 10Y(s) - sy(0^-) - y'(0^-) - 7y(0^-) &= X(s) \end{aligned} \quad \text{--- } \textcircled{A}$$

Using the given initial conditions

$$\begin{aligned} s^2 Y(s) + 7sY(s) + 10Y(s) - 2 &= X(s) \\ s^2 Y(s) + 7sY(s) + 10Y(s) &= X(s) + 2 \end{aligned}$$

$$Y(s)(s^2 + 7s + 10) = X(s) + 2 \quad \text{--- } \textcircled{B}$$

Also,

$$\begin{aligned} x(t) &\stackrel{\mathcal{L}}{\leftrightarrow} X(s) \\ e^{-jt}u(t) &\stackrel{\mathcal{L}}{\leftrightarrow} \frac{1}{s+j} \end{aligned}$$

Substituting $X(s) = \frac{1}{s+j}$, $\textcircled{C}$

$$Y(s)(s^2 + 7s + 10) = \frac{1}{s+j} + 2 = \frac{1 + 2s + 2j}{(s+j)} = \frac{2s + 1 + 2j}{(s+j)}$$

$$Y(s) = \frac{2s + 1 + 2j}{(s+j)(s^2 + 7s + 10)} = \frac{2s + 1 + 2j}{(s+j)(s+5)(s+2)}$$

$$Y(s) = \frac{2s + 1 + 2j}{(s+j)(s+5)(s+2)} \quad \text{--- } \textcircled{D}$$

1 point for trying. 1 point for A-D

- b. (5 points) Find the modes of the system represented by the above LCCDE. Calculate the forced response of this system in the Laplace domain.

The modes of the system correspond to the roots of the characteristic equation. The characteristic equation is: $s^2 + 7s + 10 = 0$ or $(s + 5)(s + 2) = 0$ or $s = -5, -2$

(A)

(B)

The partial fraction expansion can be written as

$$Y(s) = \frac{2s + 1 + 2j}{(s + j)(s + 5)(s + 2)} = \frac{A}{s + j} + \frac{B}{s + 5} + \frac{C}{s + 2}$$

The natural response corresponds to the modes of the system.

The forced response corresponds to the other term, namely

$$Y_{forced}(s) = \frac{A}{s + j}$$

(C)

Finding A,

$$A = \frac{1}{(5 - j)(2 - j)}$$

$$Y_{forced}(s) = \frac{1}{(5 - j)(2 - j)(s + j)}$$

(D)

1 point for trying. 1 point for A-D.

- c. (5 points) Calculate the final value $y(\infty)$ of this system's response.

To find the final value, we typically use the final value theorem.

However, one of the poles of the system $\frac{A}{s + j}$ is on the imaginary axis.

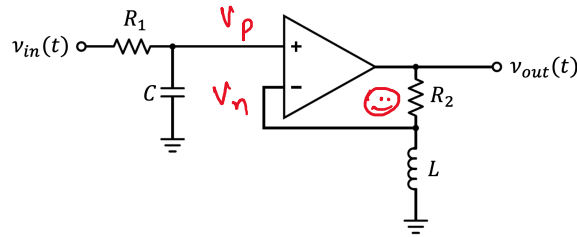
(A)

This means that the signal does not asymptotically converge to a final fixed value. It is oscillatory as $t \rightarrow \infty$.

(B)

1 point for trying. 2 points for A-B.

5. (10 points) Operational Amplifiers



a. (5 points) Calculate the transfer function $\mathbf{H(s)}$ for this circuit.

$$v_p = v_n \Rightarrow \frac{1/sC}{R_1 + 1/sC} \times v_{in} = \frac{sL}{R_2 + sL} v_{out} \quad \textcircled{A}$$

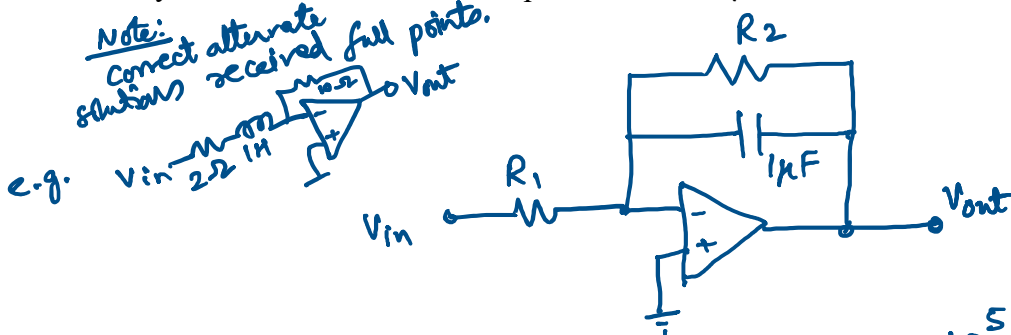
$$\Rightarrow H(s) = \frac{v_{out}(s)}{v_{in}(s)} = \frac{1}{1 + sCR_1} \times \frac{(R_2 + sL)}{sL} = \frac{R_2}{sL(1 + sCR_1)} + \frac{1}{1 + sCR_1}$$

Grading

- 1 pt attempt
- +1 pt $\textcircled{A}$
- 5 pts correct $H(s)$

b. (5 points) Sketch an op-amp circuit that implements the transfer function $\mathbf{H(s)} = \frac{-10}{s+2}$. Any capacitor used in your circuit should have a capacitance $C = 1\mu F$.

Note:
Correct alternate solutions received full points.



single-pole circuit

$$G = -10 = \frac{-1}{R_1 \times 10^{-6}} \Rightarrow R_1 = 10^5 \Omega = 100k\Omega$$

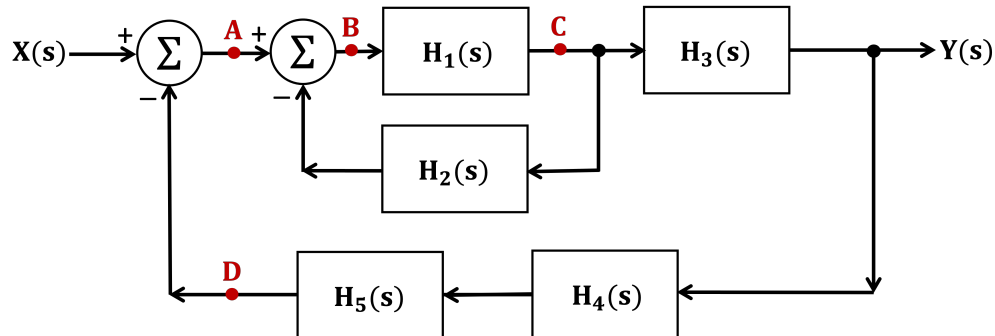
$$P = -2 = \frac{-1}{R_2 \times 10^{-6}} \Rightarrow R_2 = 500k\Omega$$

Grading

- 1 pt attempt
- +1 pt correct circuit
- 5 pts fully correct answer

6. (10 points) System Block Diagrams

Calculate the overall transfer function $\mathbf{H(s)}$ for the following connection of LTI systems. Your final answer should be expressed in the Laplace domain.



$$\begin{aligned}
 A &= X - D & - \textcircled{A} \\
 B &= A - CH_2 & - \textcircled{B} \\
 C &= BH_1 & - \textcircled{C} \\
 Y &= CH_3 & - \textcircled{D} \\
 D &= YH_4H_5 & - \textcircled{E}
 \end{aligned}$$

Substituting and simplifying

$$\begin{aligned}
 C &= AH_1 - CH_1H_2 \\
 C &= A \frac{H_1}{1 + H_1H_2} \\
 A &= X - YH_4H_5 \\
 C &= (X - YH_4H_5) \frac{H_1}{1 + H_1H_2} \\
 Y &= (X - YH_4H_5) \frac{H_1}{1 + H_1H_2} H_3 \\
 Y(1 + H_1H_2) + YH_4H_5H_1H_3 &= XH_1H_3 \\
 Y &= X \frac{H_1H_3}{(1 + H_1H_2) + H_4H_5H_1H_3} \\
 \frac{Y}{X} = H &= \frac{H_1H_3}{(1 + H_1H_2) + H_4H_5H_1H_3} \quad - \textcircled{F}
 \end{aligned}$$

$\underbrace{\hspace{10em}}_{\textcircled{G}} \quad \underbrace{\hspace{10em}}_{\textcircled{H}}$

2 points for trying. 1 point for A-H.